SAFARI SST Analysis Report

1 Full SST Time Series

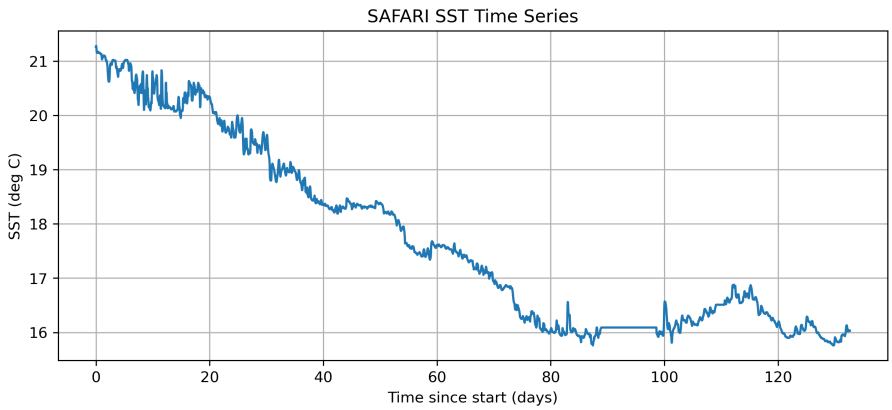

Figure 1: Full SST Time Series

The SST record covers about 133 days, from late November 2025 to early April 2026. SST decreases from about 21.2°C to about 16°C during this time, so the overall trend is cooling.

The record also shows variability at shorter timescales. There are fluctuations lasting several days to a few weeks on top of the overall cooling trend.

I plotted the full SST record first before calculating the spectrum. This made it easier to separate the long-term cooling trend from the shorter fluctuations. Looking at the time series first also helped identify which timescales were likely to appear in the PSD.

2 Spectral Analysis Method

I used 4 chunks for the PSD calculation. This gave segments about 33 days long and a frequency resolution of about 0.030 cpd.

I chose 4 chunks as a compromise between resolution and noise. Using fewer chunks would give better frequency resolution, but the spectrum would be much noisier. Using more chunks would make the spectrum smoother, but it would smear out narrow peaks and make it harder to identify the true dominant frequency.

3 Main Spectral Features

The strongest peak in the higher-frequency part of the spectrum occurs at 1.09 cpd. This corresponds to a period of about 0.92 days, or about 22 hours. This is close to the daily cycle, so it is most likely caused by daytime warming and nighttime cooling of the ocean surface.

This peak is important in the SAFARI context because the daily SST cycle can influence air-sea heat fluxes, near-surface stability, and the timing of convection. A warmer ocean surface during the day can increase evaporation and moisture flux into the atmosphere, while cooler SST at night can weaken those effects.

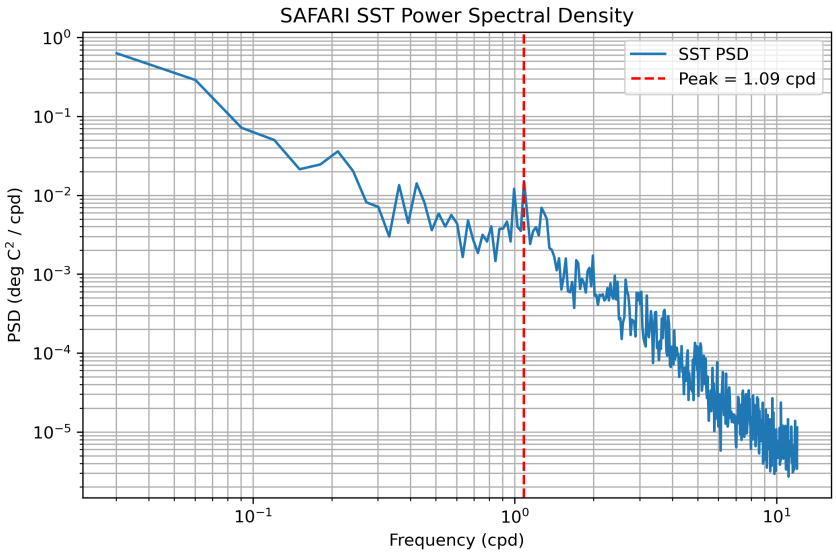

Figure 2: SST Power Spectral Density

The spectrum also has much more energy at low frequencies below about 0.1 cpd, corresponding to timescales of roughly 10–30 days. This means the largest SST changes happen over several days to a few weeks, probably because of changing weather systems, winds, and upper-ocean mixing.

4 Physical Interpretation for SAFARI

These slower SST changes are likely important for the SAFARI project because they can affect how much heat and moisture move from the ocean into the atmosphere and may influence storm development.

Overall, the two main spectral features are the strong low-frequency variability at 10–30 day timescales and the smaller peak near 1.09 cpd. The low-frequency variability probably has the larger effect on the SAFARI campaign because it changes the background ocean state for many days at a time. The near-daily peak is smaller, but it still matters because it controls the repeating cycle of daytime warming and nighttime cooling.